ROHAN DHUNDE

AI / ML Engineer | Generative AI | Agentic AI | NLP

Pune, India | +91 95886 89602 | rohandhunde5@gmail.com
linkedin.com/in/rohandhunde | github.com/rohandhunde

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying production Generative AI, Agentic AI, and traditional ML systems. Skilled in RAG architectures, LLM fine-tuning and serving, multimodal AI (OCR and speech-to-text), and FastAPI backends. Experienced in LLM evaluation, access control, and prompt-injection defense for secure, reliable AI products.

TECHNICAL SKILLS

Languages & Frameworks: Python, Async Python (asyncio), SQL, FastAPI, LangChain, LangGraph

ML / DL: PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Hugging Face, Transformers, CNN, RNN, YOLO, Computer Vision, NLP, Transfer Learning

Classical ML: Regression, classification, SVM, Naive Bayes, Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost, clustering, PCA, regularization, cross-validation, GridSearchCV

Generative AI: LLMs, RAG, LoRA/QLoRA fine-tuning, prompt engineering, OpenAI API, MCP, STT, TTS, OCR (PyTesseract)

Data: FAISS, Pinecone, Weaviate, Milvus, PostgreSQL, MySQL, ClickHouse, data modeling, data warehousing

Evaluation & Safety: LangSmith, Ragas, hallucination mitigation, prompt-injection defense, RBAC

Cloud & Delivery: AWS (EC2, S3, Lambda, RDS), Docker, CI/CD, REST API deployment, Git, GitHub

Analytics & BI: Power BI, Tableau, Matplotlib, Seaborn, statistical analysis

PROFESSIONAL EXPERIENCE

AI/ML Developer | THS Systems **Apr 2025 - Present**

- Designed and deployed RAG systems on FAISS, Pinecone, Weaviate, and Milvus for knowledge retrieval, evaluation, and automated scheduling.
- Built a multimodal RAG data lake for PDFs, spreadsheets, audio, and images, unifying SQL and vector search under RBAC-enforced access control.
- Developed an OCR + RAG + LLM pipeline that evaluates handwritten answer sheets against SQL-stored rubrics using MultiQuery retrieval and grounded validation.
- Hardened LLM-facing endpoints against direct and indirect prompt injection using input/output filtering, scoped tools, and RBAC.
- Built a WebSocket-based conversation intelligence platform for meeting minutes and clinical summaries through a configurable LLM API.
- Automated SQL-to-government-portal data entry with AI-driven Playwright MCP browser agents.
- Built an HRMS resume-shortlisting system using semantic embeddings and LLM-based scoring, plus FastAPI backends for production AI applications.

Data Scientist | 4ibiz Technosolution **Sep 2023 - Apr 2025**

- Developed generative and traditional AI models, including LoRA/QLoRA fine-tuning and RAG workflows, to automate business processes.
- Delivered analysis from large AWS RDS datasets through Python, Excel, and embedded Power BI dashboards.
- Automated customer support with AI chatbots to improve consistency across recurring workflows.

SELECTED PROJECTS

Applied AI systems built around real workflows

RAG | Multimodal AI | Agentic Automation | Healthcare AI | Document Intelligence

Multimodal Data Lake & RAG System
Ingestion, chunking, embeddings, and semantic retrieval across PDF, Excel, CSV, presentations, email, audio, and imagery, with natural-language SQL, RBAC, response validation, and vector lifecycle management.

MSBD - AI-Powered WhatsApp Teleconsultation
WhatsApp-based registration, AI symptom understanding from text or voice, specialist matching, appointment scheduling, remote consultation support, and two-way medical document sharing.

Civic Issue Management & Complaint Routing
Multimodal complaint classification and information extraction with automatic routing to responsible civic departments.

Handwritten Answer Sheet Evaluation
OCR-to-retrieval-to-LLM evaluation workflow for MCQ, short-answer, and long-answer formats with marks assignment.

Real-Time Conversation Intelligence
WebSocket-based transcription pipeline for meeting minutes, action items, and structured doctor-patient summaries.

Government Portal Automation
AI-driven Playwright MCP workflows that map SQL data to government portal navigation, forms, and submissions.

Document Scanning & Information Extraction
PyTesseract pipeline for Aadhaar, PAN, driving licences, passports, and admit cards using layout-specific PSM modes.

Help Desk Chatbot
RAG chatbot over scraped website data in FAISS with grounded answers and source-URL attribution.

Career & Stream Recommendation
Personalized Science, Commerce, or Arts/Humanities recommendations based on academics, interests, preferences, and career goals.

EDUCATION

Bachelor of Computer Science
University of Amaravati, India | 2019 - 2022 | CGPA: 7.9

Portfolio: portfolioproject-coral-xi.vercel.app